PRODUCT USER MANUAL

For OMI_EXTREME_MHW_ARCTIC

Issue: 1.1

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GLOSSARY AND ABBREVIATIONS

PUM	Product User Manual
OMI	Ocean Monitoring Indicator
MHW	Marine Heatwave
NetCDF	Network Common Data Form
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

Summary

This is the product user manual (PUM) for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_EXTREME_MHW_ARCTIC_area_averaged_anomalies. It contains the time series of daily index for surface and bottom marine heatwaves in the Barents Sea. The index is dimensionless and represents the area-averaged modelled temperature in the Barents Sea normalized relative to the seasonally varying 90th percentile of the temperature.

OMI_EXTREME_MHW_ARCTIC_area_averaged_anomalies product contains one dataset:

- **cmems_mod_arc_phy_temp_my_mhw_barentssea_P1D-m** contain daily MHW index for the Barents Sea.

The time series is computed using the TOPAZ4b reanalysis from the Arctic MFC (product name ARCTIC_MULTIYEAR_PHY_002_003).

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is described in the Quality Information Document (QUID) CMEMS-OMI-QUID-EXTREME-MHW-ARCTIC (see References).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_EXTREME_MHW_ARCTIC_area_averaged_anomalies
Geographical coverage	Barents Sea (68°N - 80°N; 18°E - 55°E)
Variables	- MHW_index_surface - MHW_index_bottom - Temperature_surface - Temperature_bottom
Available time series	1993-01-01 to 2024-12-31
Temporal resolution	Daily means
Horizontal resolution	N/A
Integration depth (for example)	Surface and bottom
Format	NetCDF4 CF1.7

b) Production System Description

This product provides the modelled time series of marine heatwave index for the surface and the bottom in the Barents Sea, estimated from reanalysis.

c) Dataset details

cmems_mod_arc_phy_temp_my_mhw_barentssea_P1D-m contains the daily timeseries of MHW index for the surface and the bottom in the Barents Sea.
MHW_index_surface [value day^{-1}] Marine heatwave index at the surface of the Barents Sea MHWsurface **MHW_index_bottom [value day^{-1}]** Marine heatwave index at the bottom of the Barents Sea MHWbottom **temperature_surface [Degree Celsius]** Temperature at the surface of the Barents Sea Temp **temperature_bottom [Degree Celsius]** Temperature at the bottom of the Barents Sea Temp

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) REFERENCES

See the Quality Information Document (QUID):
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-EXTREME_MHW_ARCTIC.pdf